ECE444 PRA7 Report

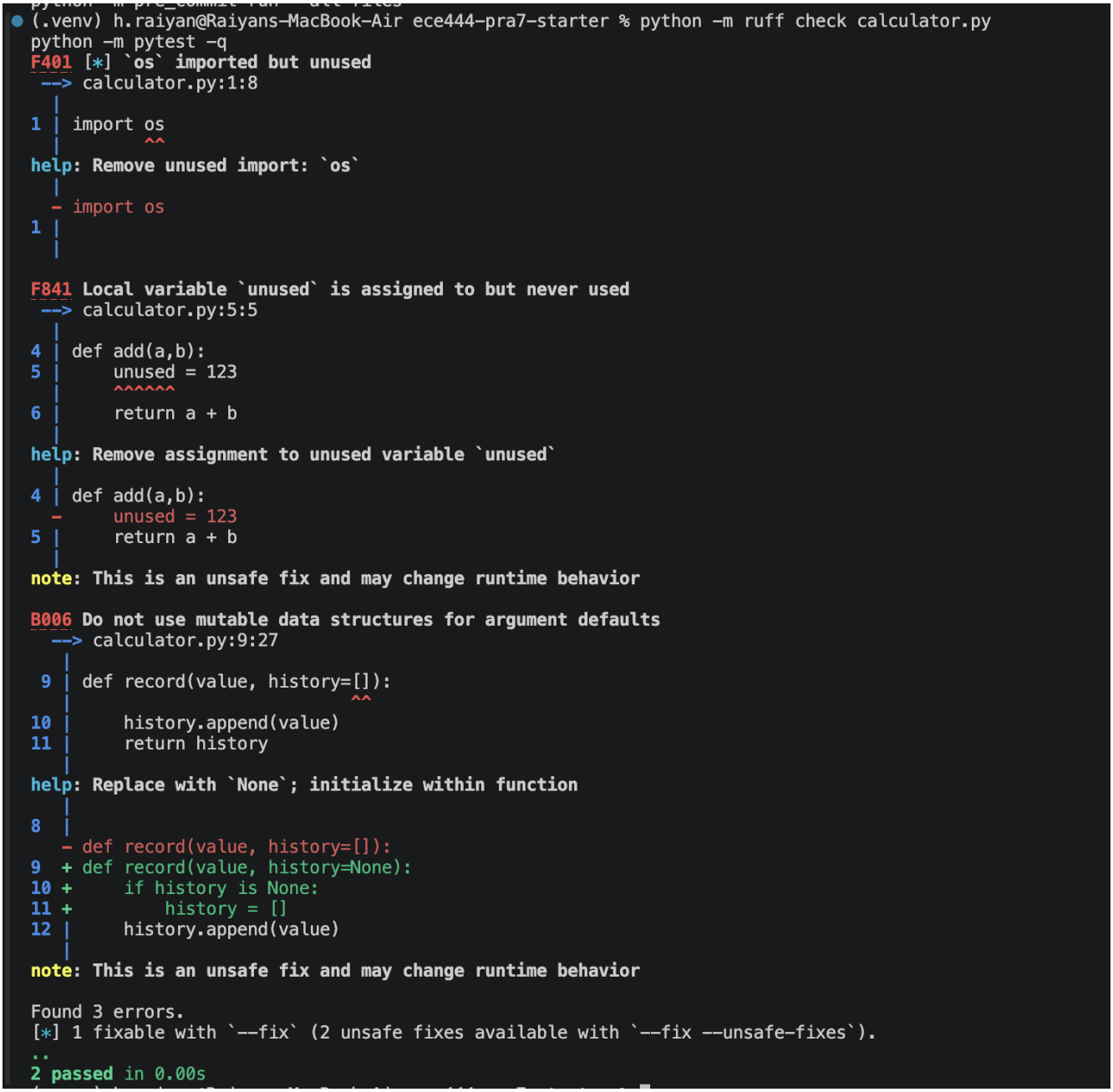
Item 1: Fork Repository URL

https://github.com/raiyan37/ece444-pra7-starter

Item 2: Ruff Check Output

Command: `python -m ruff check calculator.py`

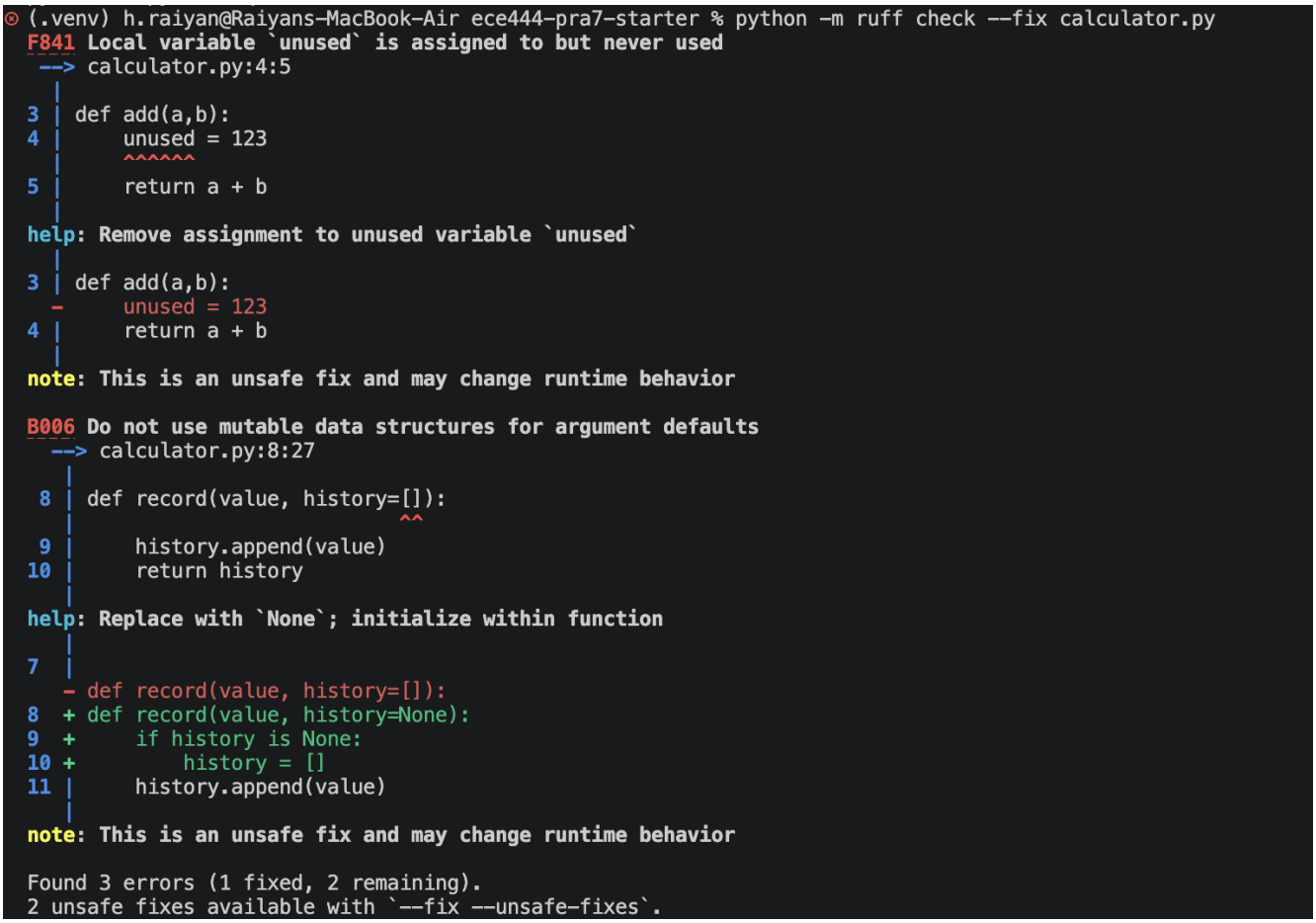
Findings reported: **F401** (`os` imported but unused), **F841** (local variable `unused` assigned but never used), and **B006** (mutable default argument `history=[]`).



Item 3: Ruff Automatic Fix Output

Command: `python -m ruff check --fix calculator.py`

Ruff automatically fixed **F401** by removing the unused `import os` (1 fixed). **F841** and **B006** remain (2 remaining) because their fixes are marked unsafe and are only applied with `--unsafe-fixes`.



```
(.venv) h.raiyan@Raiyans-MacBook-Air ece444-pra7-starter % python -m ruff check --fix calculator.py
F841 Local variable `unused` is assigned to but never used
 --> calculator.py:4:5
  |
3 | def add(a,b):
4 |     unused = 123
  |     ^^^^^^
5 |     return a + b
  |
help: Remove assignment to unused variable `unused`
  |
3 | def add(a,b):
  -     unused = 123
4 |     return a + b
  |
note: This is an unsafe fix and may change runtime behavior

B006 Do not use mutable data structures for argument defaults
  --> calculator.py:8:27
   |
 8 | def record(value, history=[]):
   |                           ^^
 9 |     history.append(value)
10 |     return history
   |
help: Replace with `None`; initialize within function
   |
7  |
   - def record(value, history=[]):
8  + def record(value, history=None):
9  +     if history is None:
10 +         history = []
11 |     history.append(value)
   |
note: This is an unsafe fix and may change runtime behavior

Found 3 errors (1 fixed, 2 remaining).
2 unsafe fixes available with `--fix --unsafe-fixes`.
```

Item 4: Repaired calculator.py and Passing Lint Check

The remaining findings were resolved manually: **F841** by deleting the unused `unused = 123` assignment in `add`, and **B006** by changing the default to `history=None` and creating a new list inside `record` on each call.

Repaired `calculator.py`:

```
def add(a, b):
    return a + b


def record(value, history=None):
    if history is None:
        history = []
    history.append(value)
    return history


def safe_divide(a, b):
    if b is None:
        return None
    return a / b
```

Command: `python -m ruff check calculator.py` → `All checks passed!`



Item 5: Additional Violation — E711 (Comparison to None)

Rule: E711 (pycodestyle, enabled via `select = ["E", ...]`) flags comparisons to `None` using == or `!=`.

Example: I added a `safe_divide` function that checks `if b == None:`.

Fix: I changed the check to `if b is None:`. == calls the object's __eq__ method, which a class can override to return something unexpected. `None` is a singleton, so an identity check with `is` is the correct and reliable test. The function behaves the same for normal inputs, and Ruff now reports no findings.

Before: code

```
11
12  def safe_divide(a, b):
13      if b == None:    Comparison to `None` should be `cond is None`
14          return None
15      return a / b
16
```

Before: Ruff output (`python -m ruff check calculator.py`***)***

```
All checks passed!
(.venv) h.raiyan@Raiyans-MacBook-Air ece444-pra7-starter % python -m ruff format calculator.py
1 file reformatted
(.venv) h.raiyan@Raiyans-MacBook-Air ece444-pra7-starter % python -m ruff check calculator.py
E711 Comparison to `None` should be `cond is None`
  --> calculator.py:13:13
   |
12 | def safe_divide(a, b):
13 |     if b == None:
   |             ^^^^
14 |         return None
15 |     return a / b
   |
help: Replace with `cond is None`
   |
12 | def safe_divide(a, b):
   -     if b == None:
13 +     if b is None:
14 |         return None
   |
note: This is an unsafe fix and may change runtime behavior

Found 1 error.
1 unsafe fix available with `--fix --unsafe-fixes`.
```

After: code

```python
def safe_divide(a, b):
    if b is None:
        return None
    return a / b
```

After: Ruff output (`python -m ruff check calculator.py`***)***

```
(.venv) h.raiyan@Raiyans-MacBook-Air ece444-pra7-starter % python -m ruff check calculator.py
All checks passed!
```